A FIELD MANUAL OF SPECULATIVE INSTRUMENTATION

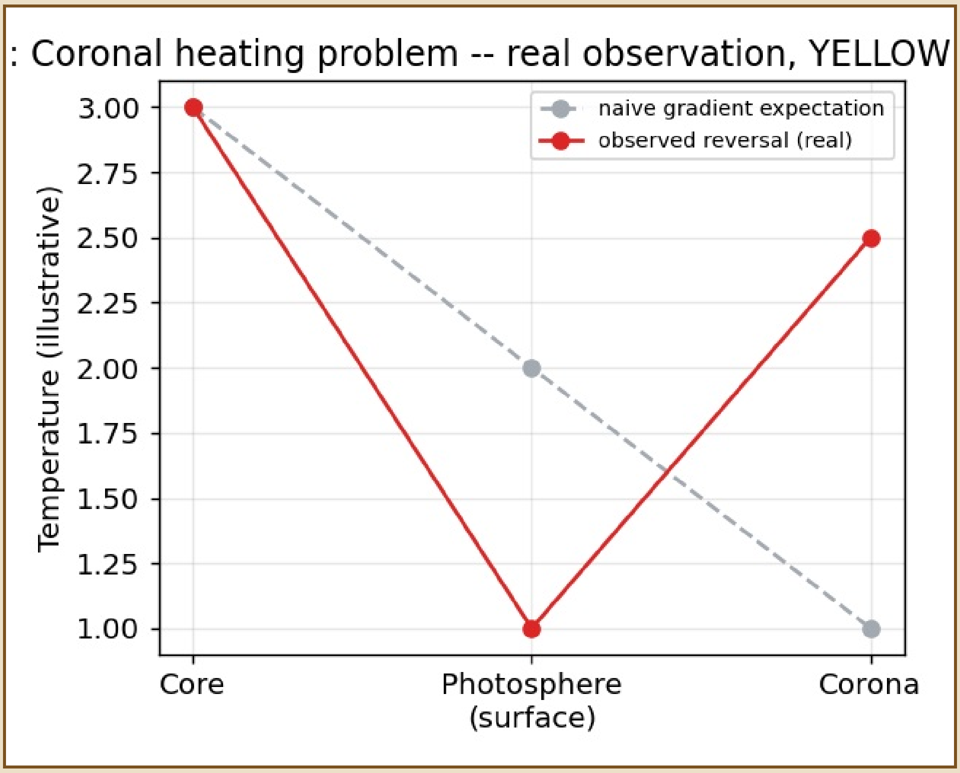
Stellar Boundary Reversal

Chapter IV — On the Sun's Own Coronal Puzzle

The sun's own corona poses an old embarrassment to simple heat-flow thinking: it is far hotter than the visible surface beneath it. Standard physics has not settled the mechanism either — magnetic reconnection, Alfven waves, turbulence all remain candidates. This chapter offers one more candidate, framed in this project's own vocabulary, without claiming to have solved what mainstream heliophysics has not.

> STANDARD REFERENCE — THE CORONAL HEATING PROBLEM
>
> Observed: core hot, photosphere cooler, corona far hotter still. The puzzle is not that the corona is hot — it is that heat appears to increase again after the visible surface, which simple conduction does not predict.

I. The Boundary as Gate, Not Final Layer



The real observed reversal (red) against the naive gradient expectation (gray).

The proposal: the photosphere is a hold boundary, not the end of heat's expression. Stored magnetic and pressure tension convert into wave release above the surface — the corona is where that stored tension is finally spent.

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interior compression -> surface hold -> magnetic/wave tension
-> boundary fold or reconnection -> outer plasma release
-> coronal heating -> solar wind
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II. Switchbacks as Mirror Events

Parker Solar Probe's observed magnetic switchbacks — brief reversals in the outward field — are proposed here as the same mirror operation this project uses elsewhere, applied to solar wind: the outward field briefly folding to carry its own opposite within it, then unfolding again.

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> No simulation yet shows a stable T_corona > T_surface condition arising from boundary energy deposition under realistic loss. The exact conversion rule from hold to release is not derived. This chapter aligns with a real, unsolved mainstream problem — it does not claim to have closed it.